

ITEC 3100 – Introduction to Networks

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Lab 2: *(Use a different color font for your written answers)*

1) Insert picture and list the functions of the following networking hardware we discussed:

a) Wired NIC- **Port for a wired ethernet**



b) Wireless NIC- **wireless wifi adapter (antenna)**



c) Cisco Switch – Connects devices, Reads MAC, Creates



tables

d) Cisco Router- Connects networks



e) Wireless access point – Acts more so as a wifi extender



f) Wireless router- Brain, combo of 3 (WAP, Router, Switch)

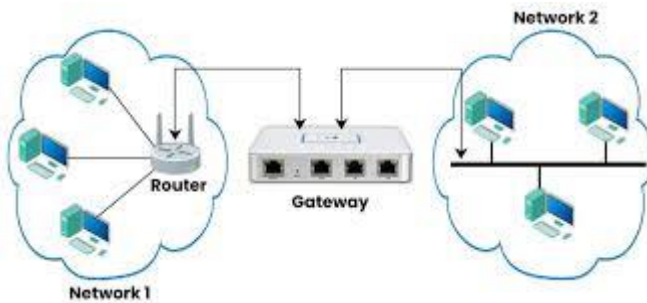


g) Network modem- translator between home network and



ISP

h) Network gateway- Combo of a wireless router and modem, point of entry and exit between networks



Watch the following video:

Hub, Switch, Router:

https://www.youtube.com/watch?v=1z0ULvg_pW8

2) What is the difference between a hub and a switch? What does it mean when we say “flooding the ports”?

Hub broadcasts to the ports

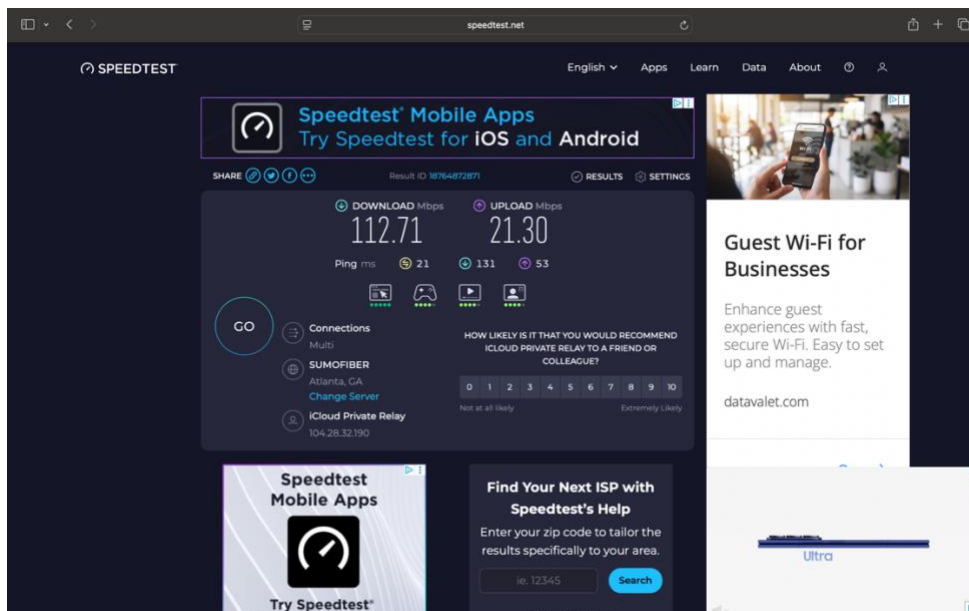
3) What is the purpose of a switch versus a router?

Switch connects devices and routers connects networks

4) Which device only understands MAC addresses? Which devices understands both MAC and IP addresses?

Network switch reads MAC, Routers understands MAC and IP

5) Go to **speedtest.net** and insert screenshot of your home WAN connection:



5) What is the difference between LAN speeds and WAN speeds? Which is faster?

LAN has less latency than a WAN because its dedicated to speed while WAN is dedicated to covering a large area. LAN is faster.

6) What is the standard wired LAN speed today?

1 Gbps (1000 mbps)

7) In a smaller network should you have more than 1 router?
Why or why not?

No because one router is enough and multiple can cause interference

8) What 3 devices is a Wireless Router comprised of?

WAP, Router, Switch

10) How is a WAP different from a Wireless Router?

WAP acts as a wifi extender while a wireless router creates and manage the network

11) What is a Network Gateway and how is that different than a Wireless Router?

A network gateway is a router and modem put together and is a point of entry and exit between networks, it is different than a router because a router is a 3 in one combination and directs data

12) Open the command prompt and run the command **ipconfig /all** then insert screenshot. (Mac users run **ifconfig**)

What are your MAC and IP addresses?

```
Last login: Fri Jan 23 19:56:12 on console
shill@macbookair ~ % ifconfig
lo0: flags=8049<UP,LOOPBACK,RUNNING,MULTICAST> mtu 16384
    options=1203<RXCSUM,TXCSUM,TXSTATUS,SW_TIMESTAMP>
    inet 127.0.0.1 netmask 0xffff0000
    inet6 ::1 prefixlen 128
    inet6 fe80::1%lo0 prefixlen 64 scopeid 0x1
    nd6 options=201<PERFORMNUD,DAD>
pf0: flags=8010<POINTOPOINT,MULTICAST> mtu 1280
    flags=0<> mtu 1280
atf0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=400<CHANNEL_ID>
    ether 26:de:81:aa:ff:7b
    media: none
    status: inactive
anp1i: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=400<CHANNEL_ID>
    ether 26:de:81:aa:ff:7c
    media: none
    status: inactive
en3: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=400<CHANNEL_ID>
    ether 26:de:81:aa:ff:5b
    nd6 options=201<PERFORMNUD,DAD>
    media: none
    status: inactive
en4: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=400<CHANNEL_ID>
    ether 26:de:81:aa:ff:5c
    nd6 options=201<PERFORMNUD,DAD>
    media: none
    status: inactive
en1: flags=8963<UP,BROADCAST,SMART,RUNNING,PROMISC,SIMPLEX,MULTICAST> mtu 1500
    options=400<TSO4,TSO6,CHANNEL_ID>
    ether 36:7b:d2:2c:66:48
    media: autoselect <full-duplex>
    status: inactive
en2: flags=8963<UP,BROADCAST,SMART,RUNNING,PROMISC,SIMPLEX,MULTICAST> mtu 1500
    options=400<TSO4,TSO6,CHANNEL_ID>
    ether 36:7b:d2:2c:66:44
    media: autoselect <full-duplex>
    status: inactive
bridge0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
    options=63<RXCSUM,TXCSUM,TSO4,TSO6>
    ether 36:7b:d2:2c:66:48
    Configuration:
        id 0:0:0:0:0:0 priority 0 hellotime 0 fwddelay 0
        maxage 0 holdcnt 0 proto stp maxaddr 100 timeout 1200
        root id 0:0:0:0:0:0 priority 0 ifcost 0 port 0
        ipfilter disabled flags 0x0
        member: en1 flags=3<LEARNING,DISCOVER>
            ifmaxaddr 0 port 0 priority 0 path cost 0
        member: en2 flags=3<LEARNING,DISCOVER>
            ifmaxaddr 0 port 0 priority 0 path cost 0
    nd6 options=201<PERFORMNUD,DAD>
    media: <unknown type>
    status: inactive
ap1: flags=8822<BROADCAST,SMART,SIMPLEX,MULTICAST> mtu 1500
    options=400<CHANNEL_ID>
    ether 22:26:92:b9:d5:92
    nd6 options=201<PERFORMNUD,DAD>
    media: autoselect (none)
en0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
```

IP- 192.168.86.30

MAC- ce:e5:23:cf:0f:ef

13) What is the purpose of a modem?

Is a translator between home network and ISP

Complete and upload to the Chapter 2 Lab dropbox.